# EXPERIMENT-11

**11. Scenario :** You are a data scientist working for a company that sells products online. You have been tasked with creating a simple plot to show the sales of a product over time.

**Question:**

1. Write code to create a simple line plot in Python using Matplotlib to predict sales happened in a month?

2. Write code to create a scatter plot in Python using Matplotlib to predict sales happened in a month?

3. Develop a Python program to create a bar plot of the monthly sales data.

**Question 11(a): Line Plot of Monthly Sales**

**Aim:** To visualize monthly sales using a line plot in Matplotlib.

**Algorithm:**

1. Import Matplotlib.
2. Define months and sales values.
3. Create a line plot using plt.plot().
4. Add labels and display the plot.

**Code:**

```
import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May']

sales = [100, 150, 120, 180, 200]

plt.plot(months, sales, marker='o')

plt.xlabel("Month")
```

plt.ylabel("Sales")

plt.title("Monthly Sales")

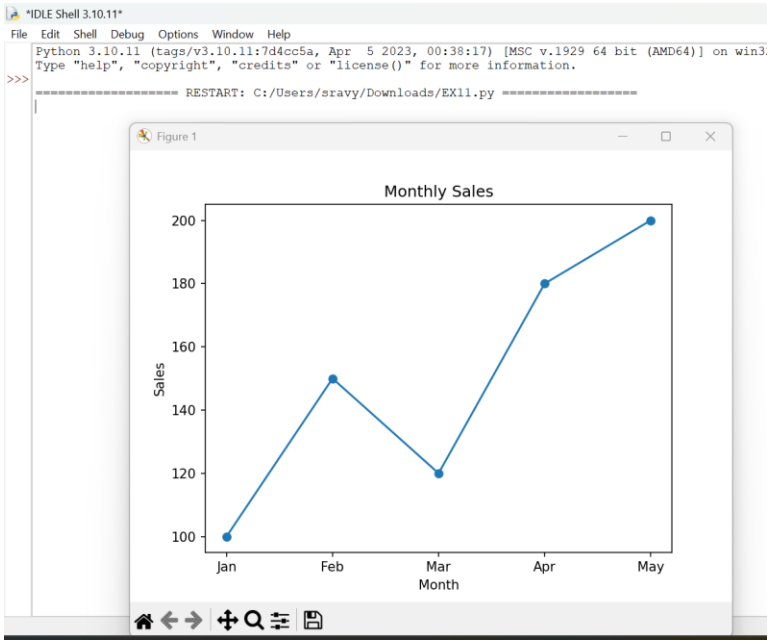
plt.show()

**Input:**

Months = Jan, Feb, Mar, Apr, May

Sales = 100, 150, 120, 180, 200

**Output:** A line plot showing monthly sales, with May having the highest sales.




**Result:** The monthly sales were visualized successfully using a line plot.

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**Question 11(b): Scatter Plot of Monthly Sales**

**Aim:** To visualize monthly sales using a scatter plot in Matplotlib.

**Algorithm:**

1. Import Matplotlib.
2. Define months and sales values.
3. Create a scatter plot using plt.scatter().
4. Add labels and display the plot.

**Code:**

```
import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May']

sales = [100, 150, 120, 180, 200]

plt.scatter(months, sales)

plt.xlabel("Month")

plt.ylabel("Sales")

plt.title("Monthly Sales Scatter Plot")

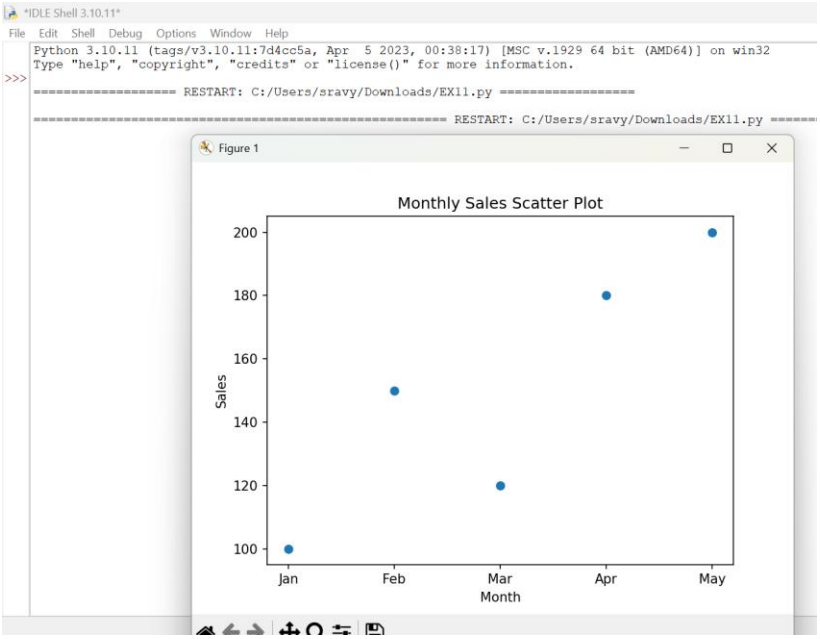
plt.show()
```

**Input:**

Months = Jan, Feb, Mar, Apr, May

Sales = 100, 150, 120, 180, 200

**Output:** A scatter plot displaying monthly sales values as individual points.




**Result:** The monthly sales were visualized successfully using a scatter plot.

**Question 11(c): Bar Plot of Monthly Sales**

**Aim:** To represent monthly sales using a bar plot in Matplotlib.

**Algorithm:**

1. Import Matplotlib.
2. Define months and sales values.
3. Create a bar plot using plt.bar().
4. Add labels and display the plot.

**Code:**

```
import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May']

sales = [100, 150, 120, 180, 200]

plt.bar(months, sales)

plt.xlabel("Month")

plt.ylabel("Sales")

plt.title("Monthly Sales Bar Plot")

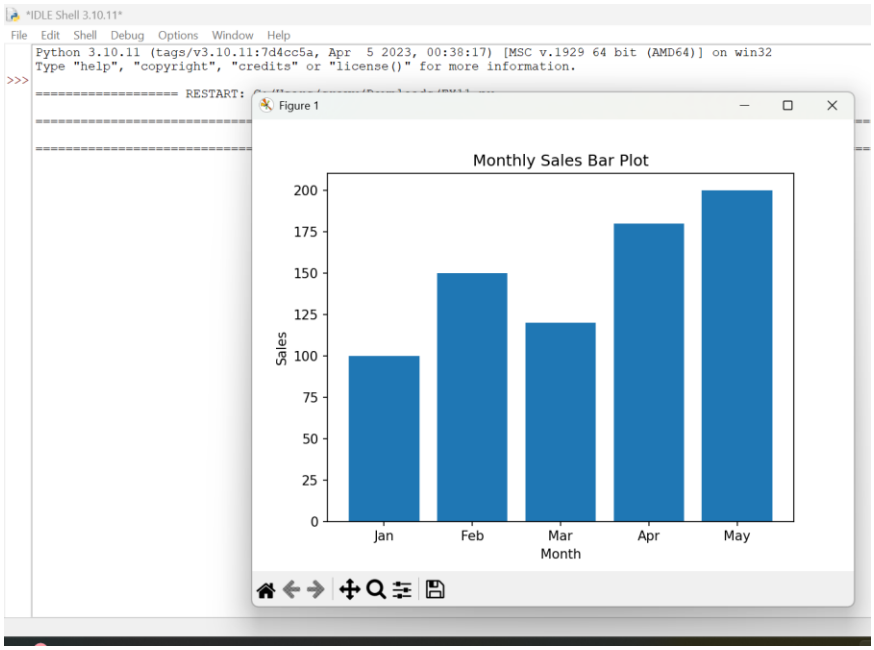
plt.show()
```

**Input:**

Months = Jan, Feb, Mar, Apr, May

Sales = 100, 150, 120, 180, 200

**Output:** A bar plot displaying monthly sales, with May having the highest sales.




**Result:** The monthly sales were represented successfully using a bar plot.